

Module 10: Cost of capital

Topic 186: Cost of Capital

We are starting a new discussion which is related to the Cost of Capital.

Capital is a term which is used commonly in accounting and finance, initially when we talk about the basic financial accounting, the term **Capital** refers to the investment of the owners of the business.

But this term Capital is used in deference ways in accounting and finance. Here the **Capital** is the amount which is invested by all the people who want to earn some return from the business.

Types of Capital:



Equity: In simple words, Equity is the investment of owners of a business and when we are talking about a corporation, a limited company or public limited company, here we have got Equity capital issued in the form of shares of the company.

So, when we issue the shares as a company to the individuals then they become **equity holders** and the funds that they provide is called the **equity capital.**

Debt: Similarly, a business organization or a corporation also generate funds or financing by issuing debt or by taking loans. Debt can be issued to the general

public in the form of bonds or debentures or it can be in the form of loan taken from a bank or a financial institution.

So, these two **Equity and Debt** are collectively called the **external types of capital** that is capital which is generated externally.

It is very important to remember that when we use the term internal or external, it is with reference to that business organization meaning that those people who are shareholders who provide equity and those people who provide loan or debt, all these people are outside the business and therefore whatever funds they provide we call it **External Capital**.

So, in **external capital** we have Debt and Equity. We also have some internal sources of capital that we will discuss later but that is the profit generated by company itself.

Cost of Capital:

Why would anybody provide funds or finances to any business organization?

It is because that they require some return from that business and that return is to be seen from two different angles.

From the businesses or corporations perspective, **this is a Cost** and this is what the business has to pay to the equity holders that we call shareholders and to the bond holders that we call **debt or loan**. And company has to pay return to these both shareholders and bond holders which is a cost or an expense for that company.

But if we see this from these investors or shareholders point of view then **it is return for them**. When we see that on one side, **it is a cost for that company** and **it is return for shareholders or bond holders**.

- Cost of Capital
- Required Rate of Return
 - Market Return
 - Opportunity Cost
 - Other Benchmarks



Required Rate of Return:

The shareholders or bond holders determine their required rate of return which is called **Cost of Capital** for that company.

Market Rate of Return: It may be the required rate of return that any investor is getting on average return in the market. And this is the easiest and common way to calculate the required rate of return and we said earlier that this becomes the cost of capital for the company.

Opportunity cost of capital: If as an individual, we have such an opportunity to invest anywhere which can earn for us more than the market rate of return then our required rate of return will become increase than market rate of return. It is because of the Opportunity Cost.

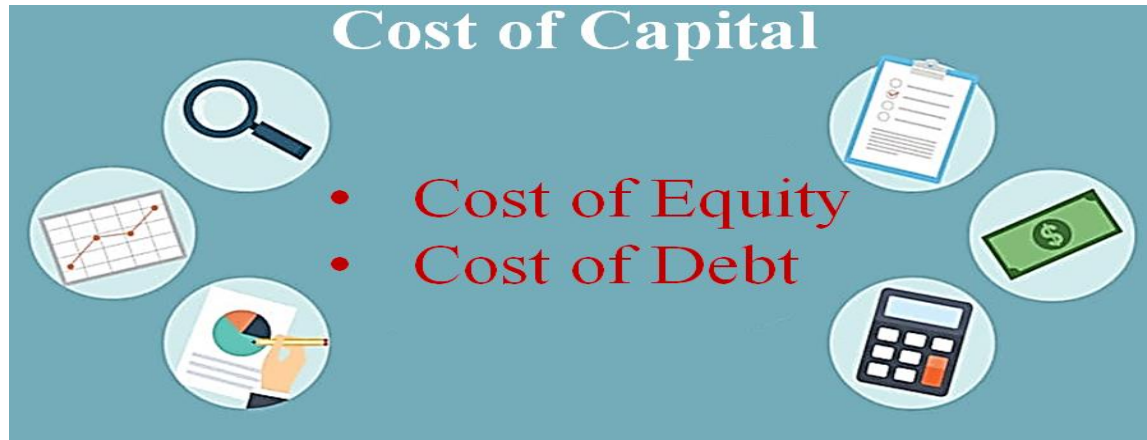
Other Benchmarks: There are other ways of calculating required rate of return. For example, we may consider the past returns earned by the company or the business meaning that in the previous or past few years how much return or profit they have been given to the shareholders or bond holders and that may be criteria.

Similarly, if we see that in this industry or sector then what rate of return has given on average which may also be made a Benchmark. So, we can say that required rate of return either will be market rate or may be an opportunity cost behind it or any industry's average or company's past record.

So, this required rate of return is then treated as the cost of capital for the company that is the company has to generate this much amount of return for their fund

providers for their capital holders to satisfy them so that they keep their investment in the company.

Let's have a look at the cost of capital in a little more detail.



Cost of Capital is related to the Equity and Debt.

Cost of Equity: The equity is a common share or preferred share. The return to the common shareholders is determined by considering the price that they are paying to buy those shares and the dividend or the profit which the company is paying to the shareholders which determines that how much return they are getting from that. So, there is an important point to remember that we do not consider the issue price of original shares at the time of determining the Cost of Equity but we do take the current market price.

Let's suppose 5 years ago, if a share was issued for Rs.10 and now that share is of Rs.50 then we will consider that share on today's price. This is because the return matters on that money or amount for shareholders which is today's value of their investment that is Rs.50 per share and not Rs.10 which was the original value. This is how we will calculate the cost of equity.

Cost of Debt: The Debt is generally termed as fixed return or fixed income capital or fixed cost of capital is there in case of debt. Although it is not true in all the cases but in most of the cases it is correct meaning that cost of debt remains constant if we relate it to the issue price.

If we are talking about bonds or debentures or debt securities which are traded in the open market where we also (like equity or shares) have to relate that return that

we pay on that debt with market value of these debt securities. In case of debts, this return is the interest that is paid on bonds or debentures or loans.

We can conclude our discussion that **Capital** which is mainly the external capital comprises of equity and debt. And the cost of equity and the cost of debt are to be separately calculated and mainly these costs are calculated by considering their market values or market prices.

And on one side for the investors, this is their required rate of return and on the other side for the company; it is their cost of capital.

Module 10: Cost of capital

Topic 187: Cost of Equity Capital

In this lecture, we will be discussing the Cost of Equity.

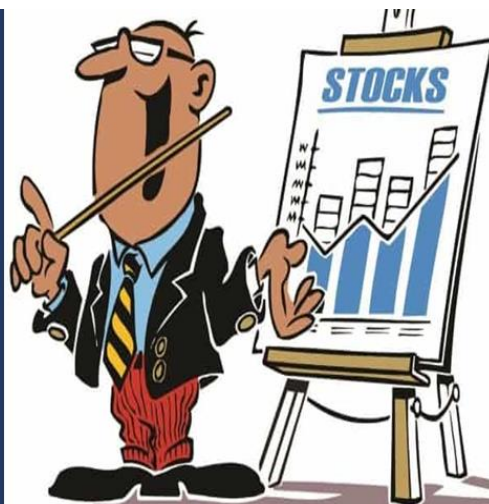
Equity:

As we know that Equity means that investment which the owners or shareholders of that company do in this company. These shareholders become owners of any company through this investment and the risk of profit and loss of that company becomes their risk.

Types of Equity:

Equity

- Common Stocks
- Preferred Stocks



Common Stocks: Equity in its true sense is actually the common stocks or common shares. There are **some characteristics of common shares** which are; common shareholder or common stock holders are real owners of any company because they actually share the risk of that business, the real profits which will be earned or the losses are actually a problem of common shareholders and in its true sense, these are the people who are the owners of a business.

They have got several **rights** i.e., once a company has issued some shares to the common shareholders and if the company wants to issue more shares then the existing common shareholders have the first right to buy these additional shares and this is done through the **Issuance of Right Shares.** They also have a right actually to participate in the decision making in the annual general meetings of the company and to vote for acceptance or rejection of certain resolutions put forward in these meetings.

They also have a right if a company goes into liquidation or wound up or if a company is finished then also that time common shareholders have right that their investment should be return even by selling the assets. But common shareholders' right on the asset is the last right that means money or funds are given to common shareholders after giving money or funds to all other stakeholders.

Preferred Stocks: The other type of Equity which may call the Equity is Preferred Stock or Preferred Shares. The **Preferred shares or Preference shares** are actually those shares where some preferences are given. These are basically **two preferences** which we discuss in the discussion of Equity; one is that when any business or company distributes dividend then firstly the dividend is given to preferred shareholder or preference shareholders and then remaining dividend is given to the common shareholders. So, because they get preference in the distribution of dividend therefore they are called **Preferred Share or Preference Shares.**

Second case is that when a business is liquidated or when it is wound up then preferred shareholders have the first right, the right before the common shares on the assets of the business meaning that they will be given money or funds before

the common shareholders. And if there will be no assets or money left after selling all assets then may be nothing will be given to the common shareholders.

So, these are some basic features of preferred shares or stocks and common shares or stocks.

Debt:



Debt is a fixed income security. Debt can be in the form of loan as we have seen that we go to any bank or financial institution and take a certain amount of loan for our business on which we have to pay some interest and have to return principal amount in a certain period of time.

Bonds: But we may also as a business borrow money directly from the people and that is by issuing the bonds.

Fixed Income Securities: These bonds are also called the Fixed Income Securities. And whoever purchased that bond and give us money in the form of finances and we against that amount pay some interest on that amount. So, the bonds are also termed as Fixed Income Securities.

Market Price and Cost of Capital:

The important element is the Market Price of bonds/common shares/preferred shares. In the discussion of Cost of Capital, we also have some discussion on it earlier that determination of the cost of bonds or shares (common or preferred), we need to consider the dividend that we are paying or the interest that we are paying and then we have to relate it to the market price of those bonds or shares.

It means that if the dividend of a particular stock suppose its constant or we suppose that if a company A's dividend is constant and it is Rs.10 per year. If the price of that stock in the market lets suppose its Rs.50 then this 10% dividend will become $(10/50 \times 100)$ 20% return to that share.

But if the price increases and in the stock market it goes up to Rs.100 then the same Rs.10 dividend will become 10% return to that shareholder who will be holding that stock at the time when the dividend was declared.

Similarly, the bonds are traded in the market and the prices of bonds also fluctuate. There are many factors that affect market prices and we are not going in detail of that on which factors the market price of any security or bond or share is dependent on but we can say that it is dependent on Demand and Supply.

So, demand and supply are the two factors which actually take the price up or down but whatever is the price, once that return which is dividend or interest link with price then it is called that Cost of Debt or Cos of Equity.

We can conclude our discussion that Preference shares are those shares which receive money or funds first in case of the distribution of dividend or in liquidation and more important thing is that rate of dividend of preference shareholders is usually constant or fixed like bonds.

But the basic difference in bonds and preference shares is that if there is profit available or not but even then interest is given to bond holders. And it is calculated and distributed before the calculation of profit.

On the other hand, whether dividend is fixed, it is paid to the preferred shareholders is still contingent on the amount of the profit earned that is even the fixed dividend on preferred shares is also paid out of the profit.

So, suppose that in any year the company does not earn any profit this means that the dividend will not be paid considering that there were no retained earnings.

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Topic 188: Weighted Average Cost of Capital WACC-1

Weighted Average Cost of Capital (WACC) is basically average cost. We know that **average** means that if there are values which are higher or lower, we add them up and divide the number of these values then we take the mean which is called the average of the values.

Average can be simple average or it may be called the weighted average.

For example, if we have 2kg of wheat whose price is Rs.50 per kg and there are also 2kg of rice and its price is Rs.100 per kg then we calculate the average of price by adding Rs.50 and Rs.100 and then divide this sum by 2 then there will be no difference.

But if quantities of rice (I.e., quantity is 5kg and price is Rs.100) and wheat (I.e., quantity is 2kg and price is Rs.50) are not equal then in order to calculate the average price, if we add up Rs.100 and Rs.50 and divide it by 2 then we can say that average price ($\frac{100+50}{2} = \frac{150}{2} = 75$) of wheat and rice is Rs.75 but it will be wrong. It means that this average will be correct if the quantities of the values whose average we are calculating should be same or in other words, we can say that their weights should be equal.

If the weights of the two items or commodities or two things that we want to average, are not the same. Then we should not calculate simple average but we should go for the weighted average.

In weighted average, we also consider their weight along with the item or the amount that needs to be averaged. Here **weight** means that it may be a quantity or any amount or its significance in that average.

For example, we want to calculate the average of cost of capital. Suppose we have got equity capital and debt capital and these two are equal meaning that we have 50% equity & 50% debt and also the cost of equity is 20% and cost of debt is 10%.

Now, our simple average and weighted average will be same meaning that if we weight (here weight is percentage of investment in the company or the percentage of the capital in the total capital of the company) is 50% for each equity and debt that means their weights are equal and as we know that if weights are equal then even the simple average would be the same as that of the weighted average.

So, if we add 10% and 20% which becomes 30% and then divided by 2 so, our weighted average or simple average will be 15% and that will be correct only if the amount of equity and debt are same. But if these two amounts are different that is the percentage of equity or debt are not same then the average we calculate in the way of simple average will not be the same as that of weighted average and we will have to calculate the weighted average.

One thing that we need to remember that we name such percentage as **Capital Structure** meaning that how much percentage of equity and debt is there in any company or corporation, we call it as a whole **Capital Structure of a company.**

So, **Weighted Average Cost of Capital (WACC)** is the average cost of equity and debt calculated on the basis of percentage of each in the total capital i.e., if equity is 70% and debt is 30% then weighted average will be different from the simple average.

Why Weighted Average Cost of Capital (WACC) is important?

Once we calculate the **Weighted Average Cost of Capital (WACC)** then we know that this is the total return or percentage of return that we have to earn from the business to satisfy our debt and equity holders. It means that if our weighed average cost let's suppose becomes 23% then it means that a business has to target at 23% profit. If that business will earn 23% then the debt and equity holders both will be satisfied because they will get their required rate of return.

Weighted Average Cost of Capital (WACC) Formula:

$$\text{WACC} = [\text{Cost of Equity} \times \% \text{ of Equity}] + [\text{Cost of Debt} \times \% \text{ of Debt} \times (1 - \text{tax rate})]$$

Here we consider the effect of income tax while calculating the cost of debt which we call Tax Shield. The **Tax Shield** is actually that benefit which we get when to bear the cost of debt meaning that tax is paid on that amount which is our profit of amount. If our profit becomes less then our tax will become less. When we pay interest on debt or when we incur cost of debt then it decreases our profit. And that decrease of profit results in income taxes saving which we call Tax Shield.

So, we do consider the savings in tax; the tax shield element while considering the cost of debt. We simply multiply the cost of debt with the percentage of debt but then we multiply this whole answer with $(1 - \text{tax rate})$.

We can conclude our discussion that **Weighted Average Cost of Capital (WACC)** is such cost which any company has to bear in order to satisfy the existing debt and equity holders.

During the calculation of that cost, we consider the weights and here **weights** means that how much percent equity and how much percent is debt and then we multiply these weights with their costs and then sum up them and in this way, we get average value that we call Weighted Average Cost of Capital (WACC).

And lastly we multiply the cost of debt, the weighted cost of debt with the expression $(1 - \text{tax rate})$ in order to reduce the cost of debt against the tax benefit that we get and the sum of it gives us the **Weighted Average Cost of Capital (WACC)**. And this Weighted Average Cost of Capital (WACC) is then used in different ways, in different decisions which are made by the company related to the investment.

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Topic 189: Weighted Average Cost of Capital WACC-2

From here, we are taking our Weighted Average Cost of Capital (WACC) that we had in the last module.

We discussed the formula of **Weighted Average Cost of Capital (WACC)** where we saw that we multiply the amount of capital or the percentage of capital which may be debt or equity with the cost of that debt or equity and then we add them up to get the overall Weighted Average Cost of Capital (WACC).

We also have to consider the element of Income Tax that is the tax shield, if we are paying a certain amount of tax so, cost of debt actually gives us that tax savings so, we multiply the cost of debt with (1- tax rate) which reduces the overall cost of debt and hence the overall cost of capital of the company.

Weighted Average Cost of Capital			
Type of Capital	Amount	Cost of Capital	Cost of Capital
	Rs.	%	Rs.
Common Stocks	300,000	22%	66,000
Preferred Stocks	200,000	18%	36,000
Loans	100,000	14%	14,000
Bonds	150,000	17%	25,500
Lease	50,000	19%	9,500
Total	800,000		151,000

Our capital of equity is common stocks and preferred stocks but our debt capital can be of different types i.e., loans, bonds, lease etc. because debt can be of any type, there are many forms and there are various factors which are related to these debts.

Weighted Average Cost of Capital

$$\text{WACC} = 151,000/800,000$$

$$\text{WACC} = 0.18875$$

$$\text{WACC} = 0.18875 \times 100 = 18.875\%$$

$$\text{WACC} = ???$$

WACC Explanation

Weighted Average Cost of Capital (WACC) is actually determine the target amount of profit or return of any company that company needs to earn how much amount of minimum profit in order to satisfy the fund providers.

If the company earns on average 18.875% profit on the entire capital of Rs. 800,000 it will be able to pay to all the capital holders.

One thing that we should keep in mind that this is not the maximum amount of return or profit that company should be seeking to earn, this is rather the **minimum amount of return or profit** from that business that company should be seeking.

Because anything more than that will increase the value of the firm meaning that if a company will earn profit or return more than 18.875% then it means that company will left with some profit even after giving their required rate of return to all the fund providers and that profit will be the retained profit; this is what we call the profit which is ploughed back or that we have reinvested in the company. This is something which will grow the company at a certain rate in the future.

This calculation and significance of Weighted Average Cost have to keep in mind that Weighted Average Cost of Capital (WACC) plays an important role when we talk about cost of capital and related to decision making.

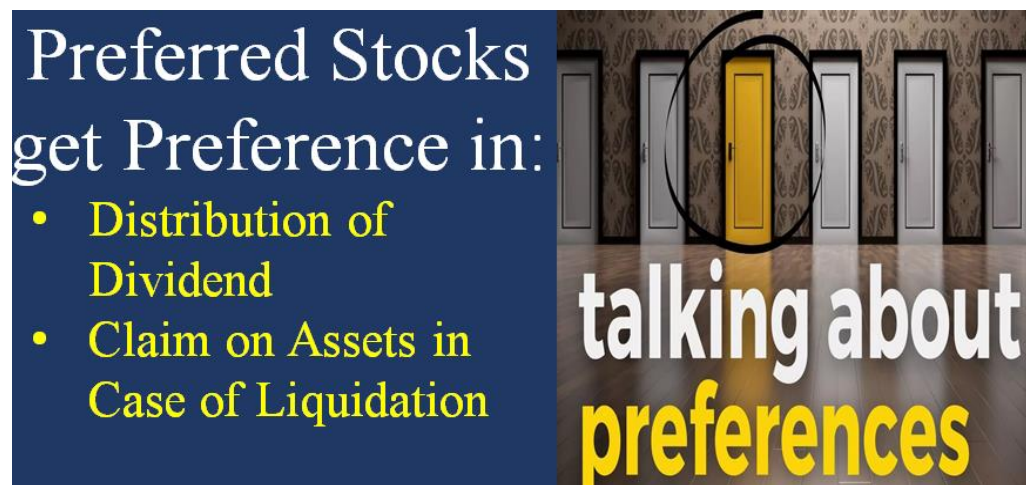
Module 10: Cost of capital

Topic 190: Cost of Preferred Stocks

In this lecture, we will discuss about the Cost of Preferred Stocks.

Cost of Preferred Stocks is calculated considering the dividend on the Cashflows which are generated by preferred stocks.

Before going in the calculation of preferred stocks, let's have a quick review about Preferred shares that what are they and what are their salient features.



Preferred shares are those stocks or shares which are preferred over common shares as: at the time of distribution of dividend and when a company is liquidated or wound up and all the money is distributed among all the stakeholders at that time they are also given preferences over common shareholders. Therefore they are called the preference or preferred shareholders.

These **preferred shareholders** take their dividend at the time of distribution of dividend and also these preference shareholders are first to take their money back at the time of liquidation. In addition to this, there are some other features of preferred shares with which these shares are issued.

For example, **they are also cumulative preference shares** meaning that the rate of dividend on the preferred shares is fixed i.e., 10% and if any year a company has not to earn 10% profit and that preference share is cumulative preference share then that 10% dividend is carried forward and then preferred shareholder will get 10% of next year and also previous year's 10% dividend with it.

Preference shares are also participating preference shares meaning that they sometimes participate with common shareholders in the remaining profit after getting their share of profit. And we call such preference shares as participative or participating preference shares.

Preference shares are also redeemable meaning that company can take back these shares after issuing them and can cancel them which can be done with common shares meaning that common shares are issued for forever. That means as long as the company exists the common shares outstanding will remain outstanding or will be issued shares but preferred shares can be redeemed or can be taken back.

Similarly, the **preference shares can also be convertible preference shares** meaning that they may be converted into common shares also.

The **cost of Preferred or Preference** shares is calculated by using a formula as:

Formula

$$r = D/P$$

- r = Cost of Preferred Stocks
- D = Preferred Dividend
- P = Market Price of Preferred Stocks

Calculations

- $r = ?$
- $D = \text{Rs. } 5$
- $P = \text{Rs. } 60$

$$r = D/P$$

$$r = 5/60$$

$$r = 8.33\%$$

We can conclude our discussion that cost of preference shareholders is determined according to the market and at that time we link amount of dividend with current market price and the percentage that we get, is called the cost of preferred shares or the cost of preferred capital.

Module 10: Cost of capital

Topic 191: Cost of Common Stocks-1

As we have discussed in earlier lectures that Common Stocks are actually those stocks or shares which provide the real ownership to the shareholders meaning that these are real owners of that company or business that bear all risk of profit and loss earned by that business. They are the true owners or the true capital providers to that company.

The return received on Common Stock is in the form of dividend and this is actually the required rate of common shareholders and for the company, it is the cost of common share capital.

Common Stocks Assumptions:

We go for three basic assumptions in order to determine the cost of common share capital because these are three possibilities which might be there when we are talking about the calculation of the cost of common stock. These assumptions are not there when we talk about the preferred or preference shares or stocks.

Because we have said that **preferred stocks'** cost is fixed in terms of percentage of dividend meaning that preferred shareholders get a fixed rate of dividend which do not fluctuate. But it is not in the case of common shareholders that every year the directors of the company sit together, they look at the profits earned or the growth of the company and they decide what are the opportunities of reinvestment of the funds and how much profit should be distributed among the shareholders.

So, the profit or dividend of common shareholders is not fixed; in some year it will be more, in some years it will be less or in some years the business or directors may decide not to pay any dividend. It means that dividend or profit of common shareholders may fluctuates because it is dependent on the amount of profit earned on one hand and on the other hand it is also dependent on the decision of the directors of the company.

By considering this, there are three assumptions which can be there and which are used to calculate the cost of common share capital.

Common Stocks Assumptions

- No Growth of Dividend
- Constant Growth of Dividend
- Growth in Stages



These three assumptions are:

No Growth of Dividend: The first assumption is that the dividend does not grow or remains constant. This is because that if any company give all the profit in the form of dividend that it has earned then that company does not grow in the upcoming years and therefore this is also no growth in the dividend.

Constant Growth of Dividend: The second assumption is that dividend is changing and this change in on which rate and we assume that amount is dividend is not constant or it is changing but the increase in rate of the dividend is constant. So, we can say that there is a constant rate of growth of dividend and we call it constant growth of dividend.

Growth in Stages: It means that the dividend of any company is not constant or it is not growing with same rate or fixed rate meaning that sometimes it increases from 5% or sometimes from 7% or sometimes decreases from 2% etc. then we call it growth in stages.

We determine the cost of common stock by doing the calculations of dividend through these three assumptions one by one.

Why no growth of dividend?

If a company which earns profit and if it retains some profit then the size of its capital's amount gets increased and next year its amount of profit will be more and growth will be seen in its profit. And if some dividend will be distributed from that profit then the growth will also be seen in dividend.

But if all the profit will be given to the shareholders in the form of dividend then shareholders will be given the constant amount of profit because **there is no growth in the company therefore dividend will not grow**. And this is the reason of no growth of dividend.

Why there is a constant growth of dividend?

If a company retains its own profit but it fixes its percentage of profit at the time of retaking that profit i.e., it will retain half profit. If a company retains its a specific

percentage of profit then in that case, the growth of the dividend will be on the constant rate.

Why growth in stages?

If the company is retaining the profit earned by it not with a constant ratio but in certain years company retains more profit and in some other years it retains lesser profit then in those cases there will be fluctuations in the amount of dividend. There will be difference in the growth of dividend like sometimes dividend will increase from 5% or sometimes from 7% or sometimes from 2% or sometimes will decrease. Then we call this situation the growth in stages that dividend gets growing but not growing constantly.

In order to understand we can make two stages of the growth of dividend and we call it two stage growth model or we can make its three stages and can call it three stage growth model and then we can extend it to any number of stages i.e., 4 stages or 5 stages or 8 stages etc. that we can make out of it.

The last important point is that every business which is not for a specific period of time or if this project's life is not determined then we apply **Going Concern assumption** on it.

The **assumption of going concern** is that the business will never be closed neither there is a requirement to shutdown that business not there is an intention to shutdown that business. And so the business will continue till infinity or in the foreseeable future. And if the business will continue in the foreseeable future then the dividend will be paid to the shareholders.

We can conclude our discussion that the amount of dividend on the common stocks is not constant because it is a matter of dividend policy of the company and the dividend policy is mainly about the percentage of distribution of profit or retention of profit that we call dividend payout or retention ratio.

So, if the retention ratio or dividend payout get changes then due to this, there arises three basic assumptions and by considering these three basics assumptions, we determine the cost of our common shares and these assumptions again are no growth of dividend, constant growth of dividend and growth of dividend in stages.

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Topic 192: Cost of Common Stocks-2

As we have discussed about common stocks that there are three basic assumptions that we can have with reference to the cost calculation of common stocks.

These three assumptions were that the dividend will never grow or will remain constant and the dividend will grow but the rate of growth will be constant so we termed it as Constant growth of dividend and third one was that there will be growth of dividend but the rate of growth will not be constant so, we talked about it may be a two stage growth model or a three stage growth model or we can extend it to any number.

No. Growth of Dividend

Formula:

$$r = D/P$$

Where 'r' is the cost, 'D' is the dividend, and 'P' is the market price.

Calculations

$$r = D/P$$

$$r = 5/50 = 10\%$$

Constant Growth of Dividend/Gordon Growth Model

Formula

$$r = D_1/P + g$$

$$\text{here } D_1 = D_0(1+g)$$

Where 'r' is the cost, 'D' is the dividend, 'P' is the market price, and 'g' is the growth rate.

Note: D_1 is actually the dividend we will get after today at the end of the upcoming year. In other words, it is the dividend of year one which will be received in the future. And when we use the term D_0 then it means that dividend which has been paid in the previous year.

Here, g is the growth rate which is a function of retention ratio and as the retention ratio will be more the same will be the growth in the coming years. Because the profit that company will retain and next year to have to earn further profit on this retained amount of profit, there will be increase in the growth of company.

Constant Growth of Dividend

Calculations

- $r = D_1/P + g$
- $r = 5.5/50 + 6\% = 17\%$

So, we can conclude our discussion that in case of no growth, the formula of perpetuity is applied which is simply $r = D/P$ and in case of constant growth we apply different formula and that formula is $r = D_1/P + g$ where D_1 is the next year's dividend calculated by $D_1 = D_0(1+g)$ and hence we get the percentage which is then used as the cost of equity capital and that can further be used to calculate the weighted average cost of capital.

Module 10: Cost of capital

Topic 193: Cost of Common Stocks-3

Growth in Stages:

- *Two-Stage Growth Model*
- *Three-Stage Growth Model*

Growth in Stages:

- *Discounting dividends of each stage*
- *Last stage always goes till infinity*
- *Apply constant growth formula on the last stage*
- *Last stage's dividends value should be further discounted for the years from where the last stage starts*

In previous lecture, we have discussed that common stocks' cost can be calculated with different assumptions and these three different assumptions are; no growth, constant growth and growth in stages. The first two assumptions that we have talked about in the earlier lectures are No Growth and Constant Growth and now will talk about third assumption that is Growth in Stages.

Growth in Stages:

The growth in stages means that the dividend of the common share of any company is changing or growing but not growing constantly meaning that if first year it is 10% then next year it will be more or less than 10% and similarly third year it will be more or less than 10%. But when we apply Growth in Stages assumption then we include another assumption in it that is; the dividend grows at a certain constant rate for a fixed or certain number of years.

Two Stage Growth Model:

Let's suppose we may say that for the first 5 years, the rate of growth of dividend will be 10% but after the 5th year from 6th year to onwards, the rate of growth will be 11% or 12% or 9%. In this way, we have divided the growth in two stages and its justification will be that for the first 5 years, company kept the retention ratio constant meaning that for first 5 years we suppose that company kept its retention

ratio or dividend payout same therefore growth of dividend remained constant. But after 5 years, that company has changed its retention or payout ratio and so, the cost will change because the amount of dividend will also change because of the retention ratio. As a result we have Two Stage Model.

Three Stage Growth Model:

And if we suppose that if the company kept the retention ratio constant for the first 5 years then changed it and then kept is constant for another 5 years and then changed it and so, we will have three stages and in this way we will have a **Three Stage Growth Model.**

In this case of Growth in Stages, we have Two Stage Growth Model and we have Three Stage Growth Model.

How do we calculate the cost of that common stock which is growing in stages when it is growing in two stages and when it is growing in three stages?

There are certain things that we need to keep in mind while making all these calculations: the dividend on the common stock will be paid till infinity so, it is not only for 5 or 10 or 20 years, growth in stages does not mean that there is a fixed time period for which we are considering the dividends. No, we are considering dividends from the first year to till infinity so, we are talking about D_1 , D_2 , $D_3, \dots, D_\infty$ which will be the dividend at the time of infinity.

So, let's have a look at the things that we need to keep in mind while going to the calculation of the cost of common stock when it is growing in stages.

Discounting dividends of each stage:

We have to separately calculate the cost of the two stages of the growth of dividend meaning that if common stock is growing for first 5 years at one specific rate and in second stage it is growing at another rate then we have to do calculations or discounting separately for both these stages. And then we will get the overall value of cost.

Last stage always goes till infinity:

In that case, the last stage of growth will always go till infinity that is; the cost of common stock which is growing in two stages, the second stage will always go till infinity.

It means that if the dividend of common stock is growing in two stages then second stage will be like constant growth model where we consider dividend till infinity. So, that will be the case in the second stage of Two Stage Growth Model. Or in the Three Stage Growth Model, the third stage of dividend will go till infinity.

Apply constant growth formula on that last stage:

Then we apply the formula of constant growth always on the last stage of the growth in stages model meaning that if we have 3 or 4 or 6 Stage Growth Models then we will apply the constant growth formula on the last stage of each model and we calculate every dividend individually on all stages before the last stage.

So, the calculation related to 2 or 3 or 4 Stage Growth Model is basically done in 2 or 3 or 5 stages; if there is 2-Stage Growth model then it will be in two stages and if there is 3-Stage Growth Model then it will be in three stages and so on. But the thing that we need to remember is that the last stage will always go till infinity.

Last stages' dividend value should be further discounted for the years from where the last stage starts:

As we have already discussed about Capital Budgeting in which we talked about IRR (Internal Rate of Return) where we said that we use Hit and Trial method to determine IRR and unfortunately here is also the same case that in 2 or 3-Stage Growth Model, we have to use the Hit and Trial method in order to determine the cost of any common stock.

It means that we use different values of 'r' in all the formulae of constant and individual values one by one and see that at what point or rate our future dividends becomes equal to our current price and we call it the cost of common stock which is growing in stages. And if at any rate, these future dividend and our current prices does not become equal then we use the 'Interpolation' and interpolate two different rates and calculate the rate which is middle of the road which is then considered as the cost of common stocks in case of this overall cost of capital.

Module 10: Cost of capital

Topic 194: Cost of Common Stocks For Specific Period

This is the discussion that we are continuing on the common stocks or shares which are held for a specific period of time.

Effect of a Specific period:

As we know that the common stocks are issued for infinite period of time and hence there is no specific time after which the stocks will be redeemed or will be

canceled. But an investor may decide about holding that common stock for a specific period of time after which the stock will be sold in the market and cash will be generated and investment will be returned.

The effects of holding the common stock for a specific period are:

- They will be a **buying and selling price** which means that the buying price is the investment of the person who is buying the stocks and the selling price will be the return that you will get at the time of selling this stock at the end of the holding period which may be more or less than the buying price. But the difference of the two (buying and selling price) would determine the return.
- In addition to this, they will specific/**fixed number of dividends** that the investor will receive over the holding period. Suppose a company pays annual dividends and we are holding it for 5 years so, we will receive 5 dividends.

How do we calculate cost of such a common stock which we want to hold for a specific period of time? Or how to calculate cost?

How to Calculate Cost?

- Hit and Trial Method
- Discount Dividends for the Holding Period
- Discount Expected Price at the End of Holding Period



- To calculate the cost of such a common stock which is held for a specific period is like IRR (Internal Rate of Return) and we need to follow the **Hit and Trial method**. It means that we will check for one rate and if we do not get correct answer on that rate then we will go for adjusting that rate and even till we reach at a rate on which there is our actual cost.
- We will **discount dividend for that holding period** and because all these dividends are individuals so, individually each dividend will be discounted.
- We also have to **discount that expected price at the end of the holding period.**

So, we have to calculate these two things in order to calculate that rate or cost of common stocks which is held for a specific period of time. It means that we have see the cost considering the dividend and also the price and when we get the value equal to the market price by discounting of both on any rate then we call **that rate** the **cost of common stock for a holding period.**

Module 10: Cost of capital

Topic 195: Cost of Bonds-1

The **cost of bonds** is the cost that a company has to bear for the bonds that it has issued.

What are bonds?

Public Debt: A bond is basically a debt security and that debt is a public debt where **public debt** means that if any company wants to take money in the form of

loan or debt from the people directly without involving any bank or financial intermediary then we call it bond in which a certificate of bond is issued and there are many types, forms and names.

Difference between bond and loan: The bond is different from loan in a way that the money is taken from the investor directly in the bonds but in the loan, the company gets money from the people through an intermediary bank or financial institution.

Secured and Unsecured: The bonds can be secured meaning that an asset can be secured by mortgaging it. And there are such bonds which are unsecured.

Cost of bonds:

The cost of bonds means that we determine that how much is the interest or return that a company has to pay to the bond holders.

This determination depends on that what is the price of that bond in the market and what rate of interest a company has determined on that price; is it fixed or variable or floating rate of interest.

Cost of Bond Vs Loan Interest:

If we compare the cost of bond and cost of loan then the cost of bond is usually less than the cost of loan. The **reason** is that because the cost of loan also includes the profit or return of the intermediary through which we get that debt financing. The corporate bonds will have a certain cost, the government bonds, the municipal bonds and there will be difference in the cost of all these bonds and that difference will be due to the risk profile or level. But the cost of bond will always be less than cost of loan.

Cost and Bond Rating:

Cost of bond depends on bond rating and as bond will more secure, its rating will increase and its cost will decrease because we have said that when as risk will decrease, the return will decrease with this decrease of risk.

Types of Bonds:

The basic types of bonds can be with reference to the maturity or with reference to the coupon or the interest rate.

No Maturity Bonds:

There are certain bonds which have got no maturity and these are called perpetual bonds or no maturity bonds.

Zero Coupon Bonds:

There are bonds where there is no interest rate and these are Coupon bonds.

Coupon Bonds:

There are coupon bonds on which there is a specific rate of interest and there is no maturity of these bonds.

What is discounted?

To calculate the cost of these bonds, we need to discount whatever is the return, whatever is the Cashflow which is generated by these bonds so, to discount that there is a certain rate that we need to apply. Now the rate determination is actually the calculation of the cost of bond.

Because there is a face value of the bond and there is a market value of the bond.

Whenever we calculate that cost of any bond then it will be done with the reference of market value or market price.

So, what we do is that we relate the interest on the bond with the market price and we see that how much is the cost of the bond with reference to that existing market price. In this way, we will consider all the cash inflow which a bond generates in the process of discounting.

Interest: We will discount the Interest, if a bond is issued for let's say 10 years so, the interest that we will get over the 10 years will be discounted.

Maturity value: And the maturity value if the bond has maturity value that will also be discounted.

In this way, the rate that we will get by making this discounted value equal to the market price will be that rate which we name **Cost of Bond**.

Module 10: Cost of capital

Topic 196: Cost of Bonds-2

We have talked about the types of bonds and said that **there are three basic categories of bonds are** Zero Coupon Bond, Coupon Bonds and No Maturity or Perpetual bonds and this categorization is with reference to the maturity and the rate of interest which is called **Coupon or Coupon Rate**.

Zero Coupon Bond:

The Zero Coupon bond means that a bond on which no interest is received.

The question is that **why would somebody invest in a bond where there is no interest?**

The answer is that the purchase price and the selling price of the maturity value are different and the difference between the two is the return of the investor.

No Installment: On these bonds there is no installment, no interest

Discount the maturity value: We have to discount the maturity value using a specific rate. Now that rate when we discount the maturity value which makes it equal to the current market price is actually the **Cost of Zero Coupon Bond**.

Formula to calculate the cost of Zero Coupon Bond: Present value formula of certain future amount where the future value is taken as Maturity value. As we need to calculate the value of 'r' which is the cost of bond so, we will reshape this formula as:

- $V = MV(1+r)^{-n}$

- $\text{Cost of Zero Coupon Bond} = r = (P/MV)^{-1/n} - 1$

Where 'MV' is the maturity value, 'r' is the rate or cost of bond and 'P' is the current price of the bond.

Cost of Perpetual Bond:

The cost of the bond which is issued for infinite period of time, no maturity or we call it **Perpetual Bond** is calculated just by considering the interest. Because **perpetual bonds** are those bonds which have no maturity so, there is no maturity value, the only thing that investor gets is installment or interest.

We need to do is that to discount that interest by applying the formula and the rate at which the discounted values become equal to the market, is the **cost of that Perpetual bond.**

Formula of Cost of Perpetual Bond:

This is the formula of calculating the value of such a Cashflow stream where there is no maturity. As we have to determine the value of cost or rate of perpetual bond 'r' therefore we reshape the formula of perpetuity as:

- $V = I/r$
- $r = I/P$

Where 'I' is the interest per year and 'P' is the current price.

Cost of Coupon Bond:

Coupon bond include both things as coupon or interest rate and other thing is maturity value. So, we have to discount both of these values and for this we have to follow the Hit and Trial method as we have seen in the discussion of IRR (Internal Rate of Return).

In such bonds, we have to discount interest or installments and also have to discount maturity value.

And the rate at which their (interest/installments and maturity value) combined present values become equal to the market price and that is our **cost of Coupon Bond.**

We can say that such rate which makes the present values of our interest and maturity value equal to our current market price then that is our **cost of coupon bond**.

Module 10: Cost of capital

Topic 197: Bond Yield And Yield to Maturity

We have discussed that **cost of bond** is actually that require rate of return which the bond holders want to earn from their investment of bond.

Bond Yield:

When we relate that rate with the market price of bond then we call it **Bond Yield**. So, **Bond Yield** is actually calculating the return to the bond holder with reference to the current or existing market price. And that is a price which a bond holder is paying currently at the moment.

Bond Yield

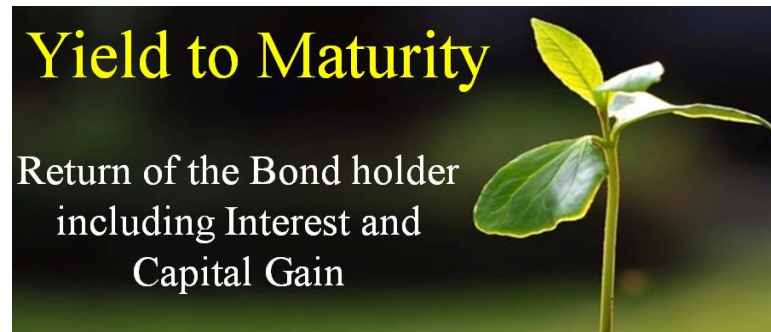
Return of the Bond holder on current market price



Yield to Maturity:

The term **Yield to maturity** also refers to the return to the bond holder with reference to the price from the beginning till the end that is the maturity period

meaning that what will be the return according to the market price till its maturity period.



So, the return of the bond holder including the interest and the capital gain is basically called the **Yield to Maturity** meaning that how much is the returns of any bond holder till his holding period or maturity period. In this, we consider both things meaning that price on which that bond has purchased and that price on which that bond has been sold and it may be its maturity value and we average it out.

Similarly, we average out the investment which we have at the start and at the end of the period.

Formula of Yield to Maturity (YTM):

$$YTM = \frac{C + \frac{FV - PV}{t}}{\frac{FV + PV}{2}}$$

Where 'YTM' is Yield to maturity, 'C' is annual interest that we get on the bond, 'FV' is price at which the bond was sold, 'PV' is the price at which the bond was purchased, 't' is time period,

When we take difference in the buying price (PV) and selling price (FV) or the initial and the maturity value and that is actually the capital gain or the profit or the loss that we get.

$$\text{YTM} = \frac{150 + \frac{1,000 - 850}{7}}{\frac{1,000 + 850}{2}} = 18.53\%$$

And when we take the difference of price at which bond was sold (FV) and the price at which the bond was purchased (PV) with the time period (t) then we get the amount of capital gain per year.

And when we add amount of capital gain per year ($\frac{FV - PV}{t}$) with interest (C) then we get total annual interest which will include capital gain as well as the bond interest.

When we divide total annual interest ($C + \frac{FV - PV}{t}$) with the value of average investment over the holding period ($\frac{FV + PV}{2}$) then we get the **Yield to Maturity.**

We can conclude our discussion that **Yield to Maturity** is actually the average return the bond holder gets over the holding period or from the beginning till the maturity of the bond which in this case comes out to be 18.53%.

Module 10: Cost of capital

Topic 198: Cost of Equity Using CAPM

We have had discussion on **CAPM (Capital Asset Pricing Model)** from different perspective earlier that we will use it for the calculation of the cost of Equity Holders. The **Equity holders** are the common shareholders, the people who are real owners of a business or a company.

Their rate of return can be calculated in different ways but one way is using the CAPM (Capital Asset Pricing Model). These are some of the important features of this model given below:

Cost of equity: This model help us to calculate the cost of equity.

Risk Free Rate: In this model, there is a consideration of Risk Free Rate of Return meaning that rate of return which any investor will receive return even if he does not take risk and this is called R_f (Risk Free Rate of Return).

Market Rate: The Market Rate of Return means that any investor can receive or will get how much return on average from the stock market which we call Market Rate of Return. We can calculate that market rate of return from the fluctuation of main index of stock market or can calculate through any biggest investment portfolio's average return.

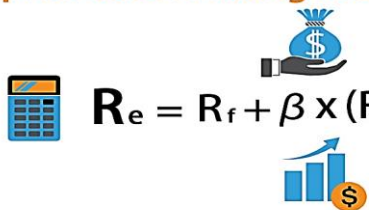
Risk Premium: There is an additional return which an equity holder or a shareholder get above the risk free rate of return and that is called Risk Premium meaning that when someone invest in the market and leave the risk free rate of return then the additional return which the investor get that is called the Risk Premium.

Beta: This Risk Premium is affected by such value which we call Index of Systematic Risk or Beta. The **Beta** is actually a measurement tool or instrument which tells us that how much risk is there in the market and is there more or less risk in a specific stock than the risk of market.

If the value of Beta will be 1 then we came to know that the return of that stock is equal to the return of market but if the value of Beta is greater than 1 that means that stock will give us higher returns. And if Beta is less than 1 that means that stock will give us lesser return.

CAPM (Capital Asset Pricing Model) Formula:

Capital Asset Pricing Model Formula



$$R_e = R_f + \beta \times (R_m - R_f)$$

Where

‘ R_e ’ is return on equity, ‘ R_f ’ is the risk free rate of return, ‘ R_m ’ is the market rate, and the difference of market rate and risk free rate of return represents the **Risk Premium** that is $(R_m - R_f)$ and anything above the risk free rate up till the market rate is the **Risk Premium** meaning that this is that additional return which shareholder will get if do investment by taking risk in the market.

And this rate of return is adjusted for what we call Beta meaning that here **Beta** means we have to multiply that value of index with beta which is telling us that risk is more or less in this particular stock as compare to the market.

A higher Beta will increase the cost of equity, a lower beta than 1 would reduce the cost of equity and beta equal to 1 will give us the exactly that cost of equity which will be equal to the market value.

So, this **CAPM (Capital Asset Pricing Model) formula** is very frequently used to calculate the cost of equity capital.

Module 10: Cost of capital

Topic 199: Economic VALUE ADDED (EVA)

Economic Value Added (EVA) is a very interesting concept and in this concept we see that a particular business or a corporation is adding to the wealth of the investors to what extent.

As we have learnt that there is a required rate of return of shareholder or bond holder and if we as investor want to earn 20% return then is business earning 20% for us or not and we will be satisfied if we are getting our required rate of return. If we are bond holder and again we will be satisfied if we get our required rate of return.

But if the business or the company will be able to generate profit more than the required rate of investor then this will be an economic addition in the size or the value or in the total wealth of that business.

So, **Economic Value Added (EVA)** is basically a measure of financial performance meaning that where we use different methods to see that performance of any company and Economic Value Added (EVA) is also one of them for measuring the performance of any business or company but this is actually a

primitive which tells us how much is our return of the business or company or the operating profit of the company in access of what is the cost of capital.

As we have discussed that **cost of capital** is actually the required rate of return of investors and when our operating profit becomes increase than our required rate of return then it means that we are further make savings even after satisfying our investors by giving them their required rate of return and this will increase the value of size of the business.

The value Economic Value Added (EVA) may be positive or negative or even zero value.

What if Economic Value Added (EVA) is negative or positive or zero?

If the value of Economic Value Added (EVA) becomes positive then it simply means that this business is earning more than expectation of investors and there is happening economic value addition.

If the value of Economic Value Added (EVA) becomes negative then it means that this business is earning less than the expectations of the investors and the size or value of this business will get shrink.

And if the value of Economic Value Added (EVA) becomes zero then it means that this business is earning exactly equal to the expectation of investors and there will be no Economic Value Added (EVA) addition.

Economic Value Added (EVA) Formula:

EVA (Economic Value Added) = NOPAT - (Invested Capital x WACC)

Where 'NOPAT' is **net operation profit after tax**, 'WACC' is **weighted average cost of capital** which is overall cost of capital and 'Invested capital' which included both debt and equity meaning that invested capital is the whole amount on which the cost is calculated.

When we multiply 'Invested capital' with 'WACC' then we get that amount which investors want to earn from our business and when this amount is deducted from the NOPAT (net operation profit after tax) then we get EVA (Economic Value Added) and if this value is positive then this means that the business is adding value.